

ACIT 1620 – Week 2 Assignment

Introduction

You have been hired as a junior web developer at a digital agency. You are tasked with supporting four different clients who each face unique website challenges:

-  BookNook – a small online bookstore
-  BeanBuzz Café – a local coffee shop expanding online
-  Tech4Kids – an educational technology startup
-  Government Health Department – a public agency improving citizen access to health information

Answer the questions carefully, complete the tables, and reflect on the scenarios presented.

BookNook's Website

 BookNook wants simple navigation for customers.

Requirement	Client Request	Your Task
Navigation Menu	"I want links for Home, Books, and Contact Us."	Identify the hyperlink type and explain why it fits. Internal links: to navigate within the website
Back to Top	"On long pages, users should quickly return to the top."	What type of hyperlink would you suggest? Anchor hyperlink: to jump to specific areas within a webpage
Author Links	"Each book should link to the author's personal website."	Explain why external links are suitable here. Takes the client to another website more about the author

- How might broken links affect the credibility of BookNook's website?

The links don't lead to anywhere and will be accounted in a website's evaluation dropping its rating.

- Do internal links influence SEO rankings? Why or why not?

Yes, as most search engines rate websites with a high amount of internal links higher when they help improve the websites navigability showing the website is professional and organized and easy to navigate for clients.

BeanBuzz Café Goes Digital

☕ BeanBuzz Café wants a new domain.

1. The owner is considering using the IP 203.0.113.25 instead of registering a name like beanbuzz.ca.

- Write down 3 benefits of using a domain name instead of an IP.

- easier to remember
- Can tell what the website is about
- Doesn't change all the time

2. They are unsure whether to use .com, .org, or .ca. Fill in the table below and recommend one.

TLD	Pros	Cons	Best Fit For
.com	Most recognized	High cost	Commercial and general purpose website
.org	Availability	Lesser recognition	Non profits
.ca	Canadian identity, trusted	Availability restrictions	Canadian businesses and organizations

- What risks might occur if BeanBuzz fails to renew its domain name?

The domain will be up for sale and no longer belong to the previous owner.

- The café wants short URLs for Instagram promotions. How do shortened links help or hurt marketing efforts? A cleaner shorter look will appear more professional and are easier to remember. Shorter links can also create trust issues as the client can't see where the link leads and could break easier.

Tech4Kids Startup

💻 Tech4Kids is preparing to launch its platform and is considering multiple domains (.edu, .org, .io). The founders want to appear credible for both schools and investors.

Tasks:

- Compare the three options and recommend the best domain strategy.
- Discuss how confusing domain choices could harm their branding.

.edu would be the best option as the website for education followed by .io as its for tech and finally but could be mistaken for a startup .org would be the worst as they wanted investors while .org is mainly used for non-profits

Write the names of each of the appropriate organization with the TLD.

TLD	Best Fit Organization
.gov	government
.edu	school
.shop	stores
.org	Non-profit
.io	Tech startups

Government Health Department

 The department wants to register a .gov domain. However, fake websites are emerging with similar names using .com.

Tasks:

- Explain the dangers of fake sites mimicking government services.
- Why should governments enforce HTTPS for all portals?

People could mistake the fake sites as the real one and enter their private information and have them stolen for malicious intent. HTTPS is an encryption protocol that encrypts communication between the client and the server stopping anyone from messing with the data when in transit.

Marketing with URLs

 BeanBuzz prints the following link on their bus ad:
https://example.com/products?id=12345&ref=bus_ad

Tasks:

- Rewrite this URL into a clean, semantic version.
 - Explain why semantic URLs are important for SEO and user trust.
- <https://example.com/products>

Helps the search engine and people understand the paths of the website better resulting in best click through rates as it's more professional and also easier to remember.

Comparison Table:

Good URL Example	Bad URL Example
https://beanbuzz.ca/menu/coffee-latte	https://beanbuzz.ca/product?id=6789&ref=ad123

Why might a company still use tracking parameters despite URLs looking less attractive?

DNS Troubleshooting

 A new domain is set up, but customers in different regions see different results.

Tasks:

- Explain what DNS propagation is.
- Estimate how long the process usually takes.
- How does browser caching influence DNS lookups?
- Why might a user in Europe see the site, but a user in Canada does not?

DNS propagation is the process when changes to a domain's DNS records are updated across all DNS servers worldwide and usually takes one to two days.

Browser caching reduces repeated lookups and speeds up page load when the browser already knows the IP address but can delay updates if the IP address changes when the old IP is still cached.

Maybe because the changes made to a DNS server haven't fully propagated to all the DNS servers around the world yet.

Case Analysis

Analyze the following domains:

- blog.education.university.ca

.ca education blog

- support.mozilla.org

.org mozilla support

- health.gov.uk

.uk gov health

Break each into TLD, SLD, and subdomains.

Now analyze these URLs:

- <https://www.shop.com/products/shoes?color=red&size=10#reviews>
- <https://developer.mozilla.org/en-US/search?q=URL>
- <https://www.health.gov.uk/vaccines/covid19?lang=en#schedule>

Fill in a table identifying scheme, domain, path, parameters, and anchor for each.

Scheme	Domain	Path	Parameters	Anchor
https	www.shop.com	/products/shoes	?color=red&size=10	#reviews
https	developer.mozilla.org	/en-US/search	?q=URL	
https	www.health.gov.uk	/vaccines/covid19	?lang=en	#schedule

Reflection & Critical Thinking

1. Why are HTTP sites insecure today?
Data is not encrypted
2. Are semantic URLs always better? Give examples.
Seo friendly, user friendly.
<https://beanbuzz.ca/menu/coffee-latte>
<https://beanbuzz.ca/product?id=6789&ref=ad123>
3. Is HTTPS enough to prove trustworthiness? Why or why not?
Yes, as it encrypts data between the client and the server.
4. What does a broken hyperlink signal about website maintenance?
That its being neglected.
5. Debate: Should companies use URL shorteners for marketing?
Yes, as its easier to remember and share but could break easier when something changes making the company seem neglected when a link doesn't work especially ones shared to a lot of people already.
6. Imagine your favorite social media site changed its domain tomorrow. How would it affect users and search engines?
The users won't be able to access the site with the old domain creating confusion, broken bookmarks and links, loss of trust, temporary drop in search rankings. 404

Submission

- A single Word or PDF file containing your answers.